



Experiment-2

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Subject Name: Computer Networks

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1. Aim/Overview of the practical:

To study and perform basic network diagnostic and network configuration commands in Windows 11, including **ping, ipconfig, tracert, nslookup, netstat, arp, hostname, pathping, route, and net use**, and understand their purpose, syntax, and practical applications in computer networking.

2. Software/Tools Required:

- **Operating System:** Windows 11
- **Command Prompt (CMD):** Built-in Windows command-line interface
- **Network Adapter:** Ethernet or Wi-Fi network connection

3. Theory:

Computer networks require various diagnostic and configuration tools to monitor communication, troubleshoot connectivity issues, and verify network settings. Windows provides several built-in command-line utilities that help users and network administrators analyze the status of network devices, identify communication problems, and obtain detailed information about network configurations.

These commands operate at different layers of the TCP/IP model. Some commands test connectivity, some display IP configuration, while others provide routing information, DNS resolution, active network connections, or address mappings between IP and MAC addresses.

These networking utilities are essential for maintaining the reliability and performance of computer networks. They enable users to quickly identify issues such as network failures, incorrect IP configurations, DNS resolution problems, routing errors, and communication delays. By analyzing the output of these commands, administrators can take appropriate corrective actions to restore normal network operation and ensure efficient data transmission.



In addition to troubleshooting, these commands are widely used during network installation, configuration, and security auditing. They provide valuable information about the current network environment, connected devices, routing paths, and system communication, making them indispensable tools for students, network engineers, system administrators, and cybersecurity professionals working with modern computer networks.

These command-line utilities are simple to use yet highly effective for diagnosing and managing network-related issues. They provide real-time information about network communication, enabling users to verify connectivity, monitor network performance, identify faults, and maintain a stable and efficient networking environment.

The commonly used networking commands include:

- **Ping** – Tests connectivity between two network devices using ICMP Echo Request and Echo Reply messages.
- **IPConfig** – Displays and manages the IP configuration of network adapters.
- **Tracert** – Shows the path (hops) taken by packets from the source to the destination.
- **Nslookup** – Queries DNS servers to obtain domain name and IP address information.
- **Netstat** – Displays active network connections, listening ports, routing tables, and network statistics.
- **ARP** – Displays and manages the Address Resolution Protocol (ARP) cache containing IP-to-MAC address mappings.
- **Hostname** – Displays the name of the current computer.
- **Pathping** – Combines the functionality of Ping and Tracert to analyze packet loss and network latency.
- **Route** – Displays and manages the IP routing table.
- **Net Use** – Displays, connects, or disconnects shared network resources.

These commands are widely used by system administrators and network engineers for troubleshooting, performance monitoring, and network management.

4.Commands Used

Command	Purpose
ping	Tests connectivity between two devices and measures response time.
ping -t	Sends continuous ping requests until stopped manually.
ping -n	Sends a specified number of ping requests.
ping -l	Specifies the packet size (payload size).
ping -f	Sends packets with the "Don't Fragment" flag set.
ping -a	Resolves an IP address to its hostname.
ipconfig	Displays basic IP configuration information.
ipconfig /all	Displays detailed TCP/IP configuration.
ipconfig /release	Releases the current DHCP-assigned IP address.
ipconfig /renew	Requests a new IP address from the DHCP server.
tracert	Displays the route taken by packets to reach the destination.
tracert -d	Displays only IP addresses without resolving hostnames.
tracert -h	Specifies the maximum number of hops.
tracert -?	Displays help for the tracert command.
nslookup	Finds the IP address of a domain name.
nslookup <IP Address>	Performs a reverse lookup (IP address to domain name).
nslookup google.com 8.8.8.8	Uses a specific DNS server for the lookup.
nslookup -type=A	Displays IPv4 (A) DNS records.
nslookup -type=AAAA	Displays IPv6 (AAAA) DNS records.

netstat	Displays active TCP connections.
netstat -a	Shows all active connections and listening ports.
netstat -n	Displays addresses and port numbers numerically.
netstat -o	Displays Process IDs (PID) for connections.
netstat -e	Displays Ethernet statistics.
netstat -r	Displays the routing table.
netstat -p TCP	Displays only TCP connections.
netstat -p UDP	Displays only UDP connections.
netstat -s	Displays protocol statistics.
netstat -ano	Displays all connections with numeric addresses and PID.
netstat -?	Displays help for the netstat command.
arp -a	Displays the ARP cache.
arp -d <IP Address>	Deletes a specific ARP cache entry.
arp -d *	Deletes all ARP cache entries.
arp -s	Creates a static ARP entry (Administrator privileges required).
hostname	Displays the computer name.
pathping	Traces the route and analyzes packet loss and latency.
route print	Displays the IP routing table.
net use	Displays or manages mapped network resources.

5.Commands output

```
C:\Users\Pranav>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=14ms TTL=114
Reply from 8.8.8.8: bytes=32 time=16ms TTL=114
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114
Reply from 8.8.8.8: bytes=32 time=10ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 16ms, Average = 13ms

C:\Users\Pranav>ping -t 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=13ms TTL=114
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 13ms, Average = 12ms
Control-C
^C
C:\Users\Pranav>ping -n 2 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=16ms TTL=114
Reply from 8.8.8.8: bytes=32 time=23ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 23ms, Average = 19ms
```

```
C:\Users\Pranav>ping -l 100 8.8.8.8

Pinging 8.8.8.8 with 100 bytes of data:
Reply from 8.8.8.8: bytes=100 time=17ms TTL=114
Request timed out.
Reply from 8.8.8.8: bytes=100 time=17ms TTL=114
Reply from 8.8.8.8: bytes=100 time=10ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 17ms, Average = 14ms

C:\Users\Pranav>ping -f -l 1472 8.8.8.8

Pinging 8.8.8.8 with 1472 bytes of data:
Reply from 8.8.8.8: bytes=1472 time=16ms TTL=114
Reply from 8.8.8.8: bytes=1472 time=24ms TTL=114
Reply from 8.8.8.8: bytes=1472 time=21ms TTL=114
Reply from 8.8.8.8: bytes=1472 time=18ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 24ms, Average = 19ms

C:\Users\Pranav>ping -a 8.8.8.8

Pinging dns.google [8.8.8.8] with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114
Reply from 8.8.8.8: bytes=32 time=17ms TTL=114
Reply from 8.8.8.8: bytes=32 time=14ms TTL=114
Reply from 8.8.8.8: bytes=32 time=14ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 17ms, Average = 14ms
```

```
C:\Users\Pranav>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi 7:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi 6:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . . :
    IPv4 Address. . . . . : 192.168.10.3
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.1

Ethernet adapter Bluetooth Network Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

C:\Users\Pranav>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-C8S4QG4
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Mixed
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
```

```
C:\Users\Pranav>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  0  7 ms  5 ms  3 ms  192.168.10.1
  1  8 ms  *  5 ms  172.20.76.1
  2  *  *  *  Request timed out.
  3  *  *  *  Request timed out.
  4  10 ms  7 ms  16 ms  47.247.118.29
  5  *  *  *  Request timed out.
  6  30 ms  10 ms  *  142.250.169.176
  7  19 ms  14 ms  16 ms  142.251.66.169
  8  16 ms  12 ms  13 ms  142.251.52.211
  9  14 ms  11 ms  10 ms  dns.google [8.8.8.8]

Trace complete.

C:\Users\Pranav>tracert -d 8.8.8.8

Tracing route to 8.8.8.8 over a maximum of 30 hops

  0  5 ms  2 ms  2 ms  192.168.10.1
  1  32 ms  *  10 ms  172.20.76.1
  2  8 ms  3 ms  3 ms  192.168.40.2
  3  12 ms  6 ms  7 ms  172.16.2.1
  4  4 ms  3 ms  8 ms  47.247.118.29
  5  *  *  *  Request timed out.
  6  12 ms  15 ms  13 ms  142.250.169.176
  7  22 ms  41 ms  12 ms  142.251.66.169
  8  8 ms  10 ms  15 ms  142.251.52.211
  9  15 ms  18 ms  15 ms  8.8.8.8

Trace complete.
```



```
C:\Users\Pranav>tracert -h 2 8.8.8.8
```

```
Tracing route to dns.google [8.8.8.8]  
over a maximum of 2 hops:
```

1	14 ms	5 ms	3 ms	192.168.10.1
2	18 ms	*	3 ms	172.20.76.1

```
Trace complete.
```

```
C:\Users\Pranav>tracert -?
```

```
Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]  
[-R] [-S srcaddr] [-4] [-6] target_name
```

```
Options:
```

-d	Do not resolve addresses to hostnames.
-h maximum_hops	Maximum number of hops to search for target.
-j host-list	Loose source route along host-list (IPv4-only).
-w timeout	Wait timeout milliseconds for each reply.
-R	Trace round-trip path (IPv6-only).
-S srcaddr	Source address to use (IPv6-only).
-4	Force using IPv4.
-6	Force using IPv6.

```
C:\Users\Pranav>nslookup 8.8.8.8
```

```
Server: DC2K16.cuchdit.in  
Address: 172.19.2.101
```

```
Name: dns.google  
Address: 8.8.8.8
```

```
C:\Users\Pranav>nslookup 8.8.8.8
```

```
Server: DC2K16.cuchdit.in  
Address: 172.19.2.101
```

```
Name: dns.google  
Address: 8.8.8.8
```

```
C:\Users\Pranav>nslookup google.com 8.8.8.8
```

```
Server: dns.google  
Address: 8.8.8.8
```

```
Non-authoritative answer:
```

```
Name: google.com  
Addresses: 2404:6800:4002:830::200e  
142.250.29.113  
142.250.29.139  
142.250.29.101  
142.250.29.102  
142.250.29.100  
142.250.29.138
```

```
C:\Users\Pranav>nslookup -type=A google.com
```

```
Server: DC2K16.cuchdit.in  
Address: 172.19.2.101
```

```
Non-authoritative answer:
```

```
Name: google.com  
Addresses: 142.250.118.102  
142.250.118.138  
142.250.118.113  
142.250.118.139  
142.250.118.100  
142.250.118.101
```

```
C:\Users\Pranav>nslookup -type=AAAA google.com
```

```
Server: DC2K16.cuchdit.in  
Address: 172.19.2.101
```

```
Non-authoritative answer:
```

```
Name: google.com  
Address: 2404:6800:4002:828::200e
```

```
C:\Users\Pranav>netstat
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49669	DESKTOP-C8S4QG4:49670	ESTABLISHED
TCP	127.0.0.1:49670	DESKTOP-C8S4QG4:49669	ESTABLISHED
TCP	127.0.0.1:49688	DESKTOP-C8S4QG4:49689	ESTABLISHED
TCP	127.0.0.1:49689	DESKTOP-C8S4QG4:49688	ESTABLISHED
TCP	127.0.0.1:49690	DESKTOP-C8S4QG4:49691	ESTABLISHED
TCP	127.0.0.1:49691	DESKTOP-C8S4QG4:49690	ESTABLISHED
TCP	192.168.10.3:49446	4.213.25.242:https	ESTABLISHED

```
^C
```

```
C:\Users\Pranav>netstat -a
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:445	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:623	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:664	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:3306	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:5040	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:5432	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:16992	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:16993	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:33060	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:49664	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:49665	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:49666	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:49667	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:49668	DESKTOP-C8S4QG4:0	LISTENING
TCP	0.0.0.0:49709	DESKTOP-C8S4QG4:0	LISTENING
TCP	127.0.0.1:8884	DESKTOP-C8S4QG4:0	LISTENING
TCP	127.0.0.1:49669	DESKTOP-C8S4QG4:49670	ESTABLISHED
TCP	127.0.0.1:49670	DESKTOP-C8S4QG4:49669	ESTABLISHED
TCP	127.0.0.1:49688	DESKTOP-C8S4QG4:49689	ESTABLISHED

```
C:\Users\Pranav>netstat -n
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49669	127.0.0.1:49670	ESTABLISHED
TCP	127.0.0.1:49670	127.0.0.1:49669	ESTABLISHED
TCP	127.0.0.1:49688	127.0.0.1:49689	ESTABLISHED
TCP	127.0.0.1:49689	127.0.0.1:49688	ESTABLISHED
TCP	127.0.0.1:49690	127.0.0.1:49691	ESTABLISHED
TCP	127.0.0.1:49691	127.0.0.1:49690	ESTABLISHED
TCP	192.168.10.3:49446	4.213.25.242:443	ESTABLISHED
TCP	192.168.10.3:50630	104.18.39.21:443	ESTABLISHED
TCP	192.168.10.3:51965	172.188.155.25:443	ESTABLISHED
TCP	192.168.10.3:53864	104.18.32.47:443	ESTABLISHED
TCP	192.168.10.3:53975	13.78.109.163:8883	ESTABLISHED
TCP	192.168.10.3:53978	23.103.238.170:443	ESTABLISHED
TCP	192.168.10.3:53980	23.103.238.170:443	ESTABLISHED
TCP	192.168.10.3:55243	35.190.80.1:443	ESTABLISHED
TCP	192.168.10.3:58750	54.192.142.37:443	TIME_WAIT
TCP	192.168.10.3:61025	52.110.14.158:443	ESTABLISHED
TCP	192.168.10.3:61153	150.171.28.11:443	ESTABLISHED
TCP	192.168.10.3:62101	150.171.28.11:443	ESTABLISHED
TCP	192.168.10.3:62786	172.64.148.235:443	ESTABLISHED

```
C:\Users\Pranav>netstat -n
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49669	127.0.0.1:49670	ESTABLISHED
TCP	127.0.0.1:49670	127.0.0.1:49669	ESTABLISHED
TCP	127.0.0.1:49688	127.0.0.1:49689	ESTABLISHED
TCP	127.0.0.1:49689	127.0.0.1:49688	ESTABLISHED
TCP	127.0.0.1:49690	127.0.0.1:49691	ESTABLISHED
TCP	127.0.0.1:49691	127.0.0.1:49690	ESTABLISHED
TCP	192.168.10.3:49446	4.213.25.242:443	ESTABLISHED
TCP	192.168.10.3:50630	104.18.39.21:443	ESTABLISHED



```
C:\Users\Pranav>netstat -e
Interface Statistics
```

	Received	Sent
Bytes	3054974060	127674540
Unicast packets	486070	585810
Non-unicast packets	1065	1485
Discards	0	0
Errors	0	0
Unknown protocols	0	

```
C:\Users\Pranav>netstat -r
```

```
=====
Interface List
9...46 a3 bb 13 fd d5 .....Intel(R) Wi-Fi 7 BE200 320MHz #7
2...44 a3 bb 13 fd d6 .....Intel(R) Wi-Fi 7 BE200 320MHz #9
23...44 a3 bb 13 fd d5 .....Intel(R) Wi-Fi 7 BE200 320MHz #6
6...44 a3 bb 13 fd d9 .....Bluetooth Device (Personal Area Network) #2
1.....Software Loopback Interface 1
=====
```

```
IPv4 Route Table
=====
```

```
Active Routes:
```

Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	192.168.10.1	192.168.10.3	45
	127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
	127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
127.255.255.255	255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
192.168.10.0	255.255.255.0	255.255.255.0	On-link	192.168.10.3	301
192.168.10.3	255.255.255.255	255.255.255.255	On-link	192.168.10.3	301
192.168.10.255	255.255.255.255	255.255.255.255	On-link	192.168.10.3	301
224.0.0.0	240.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	240.0.0.0	On-link	192.168.10.3	301
255.255.255.255	255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	255.255.255.255	On-link	192.168.10.3	301

```
=====
```

```
C:\Users\Pranav>netstat -p TCP
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49669	DESKTOP-C8S4QG4:49670	ESTABLISHED
TCP	127.0.0.1:49670	DESKTOP-C8S4QG4:49669	ESTABLISHED
TCP	127.0.0.1:49688	DESKTOP-C8S4QG4:49689	ESTABLISHED
TCP	127.0.0.1:49689	DESKTOP-C8S4QG4:49688	ESTABLISHED
TCP	127.0.0.1:49690	DESKTOP-C8S4QG4:49691	ESTABLISHED
TCP	127.0.0.1:49691	DESKTOP-C8S4QG4:49690	ESTABLISHED

```
^C
```

```
C:\Users\Pranav>netstat -p UDP
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
-------	---------------	-----------------	-------

```
C:\Users\Pranav>netstat -p UDP
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
-------	---------------	-----------------	-------

```
C:\Users\Pranav>
```

```
C:\Users\Pranav>netstat -s
```

```
IPv4 Statistics
```

Packets Received	= 1185426
Received Header Errors	= 0
Received Address Errors	= 4
Datagrams Forwarded	= 0
Unknown Protocols Received	= 0
Received Packets Discarded	= 1232
Received Packets Delivered	= 1190204
Output Requests	= 174027
Routing Discards	= 0

```
C:\Users\Pranav>netstat -ano
```

```
Active Connections
```

Proto	Local Address	Foreign Address
TCP	0.0.0.0:135	0.0.0.0:0
TCP	0.0.0.0:445	0.0.0.0:0
TCP	0.0.0.0:623	0.0.0.0:0
TCP	0.0.0.0:664	0.0.0.0:0
TCP	0.0.0.0:3306	0.0.0.0:0
TCP	0.0.0.0:5040	0.0.0.0:0
TCP	0.0.0.0:5432	0.0.0.0:0
TCP	0.0.0.0:16992	0.0.0.0:0
TCP	0.0.0.0:16993	0.0.0.0:0
TCP	0.0.0.0:33060	0.0.0.0:0
TCP	0.0.0.0:49664	0.0.0.0:0
TCP	0.0.0.0:49665	0.0.0.0:0
TCP	0.0.0.0:49666	0.0.0.0:0
TCP	0.0.0.0:49667	0.0.0.0:0
TCP	0.0.0.0:49668	0.0.0.0:0
TCP	0.0.0.0:49709	0.0.0.0:0
TCP	127.0.0.1:8884	0.0.0.0:0
TCP	127.0.0.1:49669	127.0.0.1:49670
TCP	127.0.0.1:49670	127.0.0.1:49669
TCP	127.0.0.1:49688	127.0.0.1:49689
TCP	127.0.0.1:49689	127.0.0.1:49688
TCP	127.0.0.1:49690	127.0.0.1:49691
TCP	127.0.0.1:49691	127.0.0.1:49690
TCP	192.168.10.3:139	0.0.0.0:0
TCP	192.168.10.3:49446	4.213.25.242:443
TCP	192.168.10.3:51965	172.188.155.25:443
TCP	192.168.10.3:52374	104.18.39.21:443
TCP	192.168.10.3:52376	23.217.111.160:80
TCP	192.168.10.3:52377	23.217.111.160:80



```
C:\Users\Pranav>arp -a

Interface: 192.168.10.3 --- 0x17
Internet Address      Physical Address      Type
192.168.10.1          c8-3a-35-5e-c4-98    dynamic
192.168.10.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Users\Pranav>arp -s 192.168.1.15 00-15-5D-45-12-AB
The ARP entry addition failed: The requested operation requires elevation.

C:\Users\Pranav>hostname
DESKTOP-C8S4QG4

C:\Users\Pranav>pathping 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:
 0  DESKTOP-C8S4QG4 [192.168.10.3]
 1  192.168.10.1
 2  * * * 172.20.76.1
 3  * * *
Computing statistics for 50 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
 0                                DESKTOP-C8S4QG4 [192.168.10.3]
 1    7ms     0/ 100 = 0%      0/ 100 = 0%      192.168.10.1
 2    8ms     0/ 100 = 0%      0/ 100 = 0%      172.20.76.1

Trace complete.
```

```
C:\Users\Pranav>route print
=====
Interface List
 9...46 a3 bb 13 fd d5 .....Intel(R) Wi-Fi 7 BE200 320MHz #7
 2...44 a3 bb 13 fd d6 .....Intel(R) Wi-Fi 7 BE200 320MHz #9
 23...44 a3 bb 13 fd d5 .....Intel(R) Wi-Fi 7 BE200 320MHz #6
 6...44 a3 bb 13 fd d9 .....Bluetooth Device (Personal Area Network) #2
 1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway           Interface        Metric
0.0.0.0                    0.0.0.0          192.168.10.1      192.168.10.3     45
127.0.0.0                  255.0.0.0        On-link           127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link           127.0.0.1        331
127.255.255.255            255.255.255.255  On-link           127.0.0.1        331
192.168.10.0                255.255.255.0    On-link           192.168.10.3     301
192.168.10.3                255.255.255.255  On-link           192.168.10.3     301
192.168.10.255              255.255.255.255  On-link           192.168.10.3     301
224.0.0.0                  240.0.0.0        On-link           127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link           192.168.10.3     301
255.255.255.255            255.255.255.255  On-link           127.0.0.1        331
255.255.255.255            255.255.255.255  On-link           192.168.10.3     301
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
1    331 ::1/128                On-link
23   301 fe80::/64                On-link
23   301 fe80::afe6:2f2d:b426:8c3e/128
    On-link
1    331 ff00::/8                On-link
23   301 ff00::/8                On-link
=====
Persistent Routes:
None

C:\Users\Pranav>net use
New connections will be remembered.

There are no entries in the list.
```



6.Result

The experiment was performed successfully on a Windows 11 system. Various network diagnostic and configuration commands, including **ping**, **ipconfig**, **tracert**, **nslookup**, **netstat**, **arp**, **hostname**, **pathping**, **route**, and **net use**, were executed and their outputs were observed. The experiment provided an understanding of network connectivity, IP configuration, routing, DNS resolution, active network connections, ARP cache management, and network troubleshooting techniques. Thus, the objectives of studying basic network and network configuration commands were successfully achieved.

7.Learning Outcomes

- Learned the purpose and usage of basic Windows networking commands.
- Understood how to test network connectivity and diagnose communication issues.
- Gained practical knowledge of IP configuration, DNS resolution, and routing.
- Learned to monitor network connections, ARP cache, and routing information.
- Developed hands-on skills in troubleshooting and managing network configurations using Command Prompt.